

Ex 1) A little girl is very upset because she cannot open her lunch box. In frustration, she climbs to the top of the school (9 m high) and throws her lunch box upwards with a velocity of 12 m/s. She watches her lunch box crash to the pavement below. The height of her lunch box in metres is given by $h(t) = -4.9t^2 + 12t + 9$, where t is in seconds.

a) How long does it take for the lunch box to hit the ground? [3.05 seconds]

$$h = 0 \quad 0 = -4.9t^2 + 12t + 9$$

$$t = \frac{-12 \pm \sqrt{(12)^2 - 4(-4.9)(9)}}{2(-4.9)}$$

$$2(-4.9)$$

$$t = 3.05s \text{ or } t = -0.602s$$

eg. it takes 3.05s to hit the ground

b) How high does her lunch box reach? [16.3 m]

$$\text{Avg. velocity} = +3.05 - 0.602s$$

$$= +1.209s$$

$$h(1.209) = 16.3m$$

c) If her lunch box can withstand an impact velocity of 20 m/s, will her lunch box break open so she can eat her lunch? [no]

$$h'(t) = -9.8t + 12$$

$$= -9.8(3.05) + 12$$

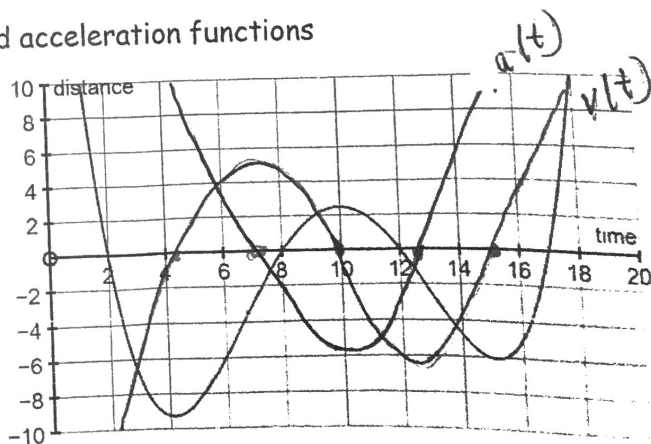
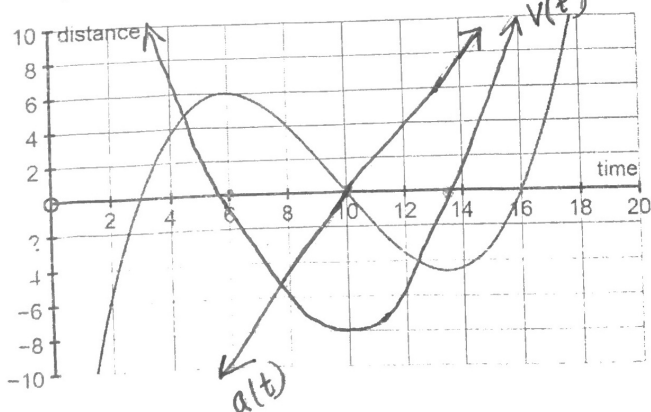
$$= -17.88 \text{ m/s}$$

eg. $17.88 < 20$, the lunch box can withstand the impact

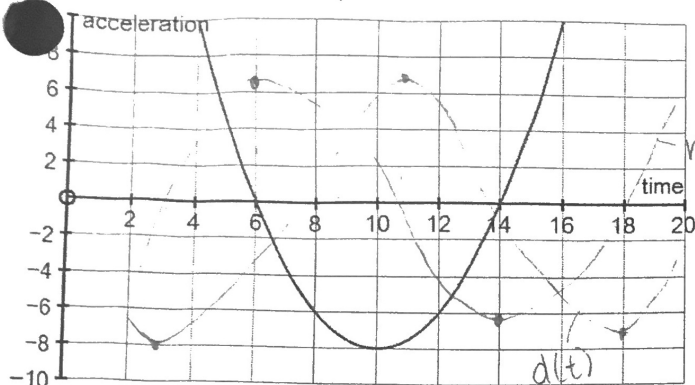
d) What is the acceleration of her lunch box?

$$h''(t) = -9.8 \text{ m/s}^2$$

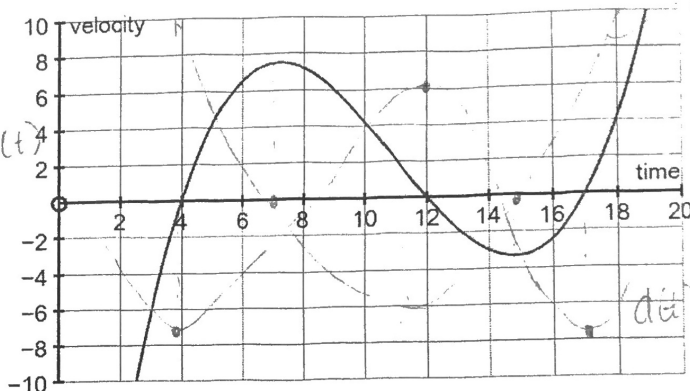
Ex) For the given graphs, sketch the velocity and acceleration functions



Ex) For the given graph, sketch the distance and velocity functions



Ex) For the given graph, sketch the distance and acceleration functions



Ex) A particle moves according to $s(t) = 2t^3 - 21t^2 + 60t - 12$ where s is in metres and t is in seconds.

a) When does the particle change direction? $[t = 2, 5]$

$$s'(t) = 0$$

$$s'(t) = 6t^2 - 42t + 60$$

$$0 = 6(t^2 - 7t + 10)$$

$$0 = 6(t-5)(t-2)$$

∴ the particle change direction at 2 & 5.

b) How quickly is the particle moving after 3 seconds? $[-12 \frac{m}{s}]$

$$s'(3) = 6(3)^2 - 42(3) + 60$$

$$= -12 \text{ m/s.}$$

c) Is the particle speeding up or slowing down at 4 seconds? $[\text{slowing down}]$

$$a(4) = 12t - 42$$

$$= 12(4) - 42$$

$$= 6$$

$$v(4) = 6(4)^2 - 42(4) + 60$$

$$= -12$$

$$v(4) \times a(4)$$

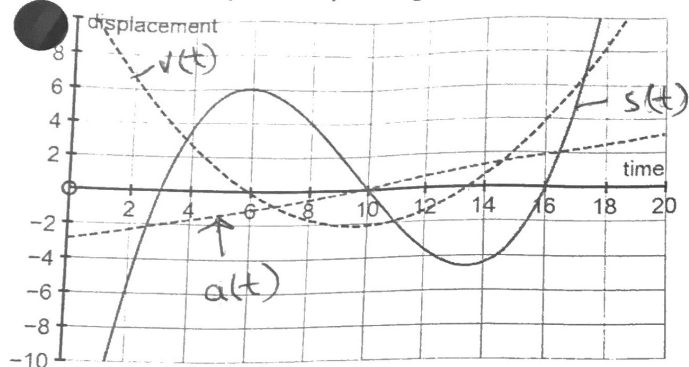
$$-12 \times 6 < 0$$

∴ slowing down

d) Sketch the motion of the particle.



Ex) Given the following displacement/velocity/acceleration time graph, state the intervals when the object is speeding up and slowing down.

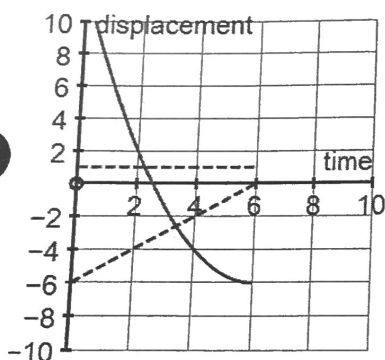


Speeding up $v(t) \times a(t) > 0$
 $x \in (6, 10) \cup (13.5, \infty)$

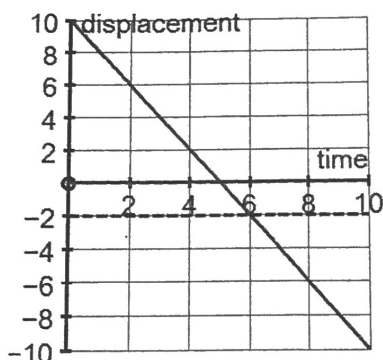
slowing down
 $x \in (-\infty, 6) \cup (10, 13.5)$

Ex) In each scenario, state

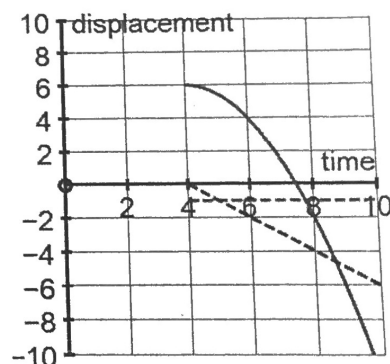
- If the function is increasing or decreasing and the sign of the velocity
- The concavity and the sign of the acceleration
- Whether the object is speeding up or slowing down.



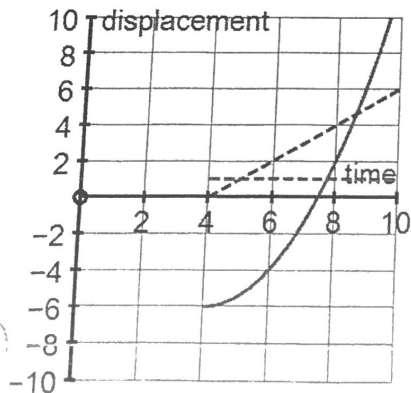
- dec, -ve
- cu, +ve
- slowing down



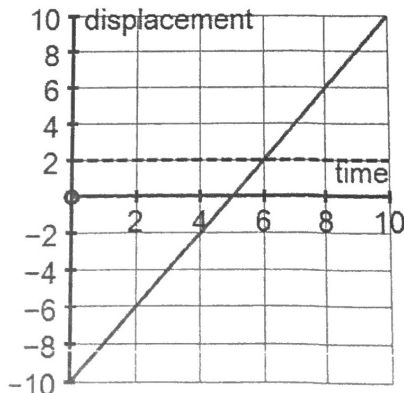
- dec, -ve
- neither, 0
- 0



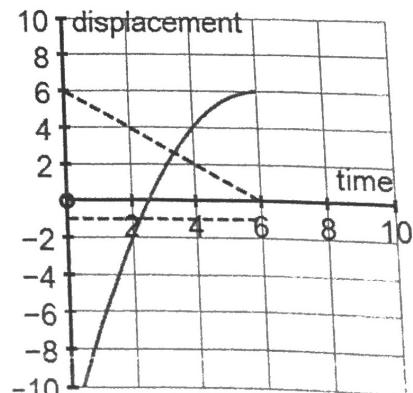
- inc, +ve
- cd, -ve
- speeding up



- inc, +ve
- cu, +ve
- speeding up



- inc, +ve
- neither, 0
- 0



- inc, +ve
- cd, -ve
- slowing down

Warm-Up

1. The position function of an object moving along a straight line is $s(t) = t^3 - 24t^2 + 144t + 15$ where t is in seconds and s is in metres. Assume that the starting position of 15 m is the reference point.

a) When does the object change direction?

$$s'(t) = 3t^2 - 48t + 144$$

$$0 = 3(t^2 - 16t + 48)$$

$$0 = 3(t - 12)(t - 4)$$

$$\therefore t = 12 \text{ or } t = 4$$

$\therefore$ the object changes direction at $t = 4\text{s}$ & $t = 12\text{s}$.

b) Is the object speeding up or slowing down at $t = 9\text{s}$

$$v(9) = 3(9)^2 - 48(9) + 144$$

$$= -45$$

$$a(t) = 6t - 48$$

$$a(9) = 54 - 48$$

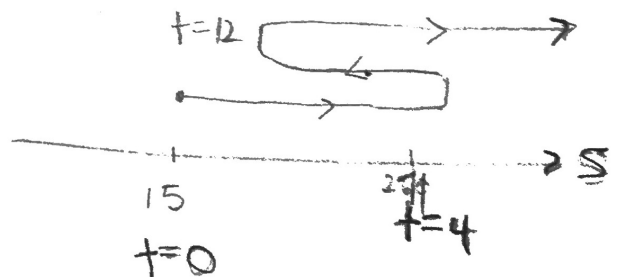
$$= 6$$

$$\therefore v(9) \times a(9) < 0,$$

$\therefore$ the object is slowing down.

Determine the displacement of the object after $t = 5\text{s}$ & sketch the motion of the particle

$$s(5) = 260\text{m}$$



The Second Derivative Test

If you are given a function, you can use the second derivative to determine if a function has a maximum or a local minimum at a critical number c , using the second derivative.

Steps:

1. Find the critical numbers of $f'(x)$ i.e. $f'(c) = 0$ or $f'(c)$ is DNE

2. Sub the critical number into $f''(x)$

• If $f''(c) > 0$ then the function has a local minimum at c .